Data Session 1 Key: Where Election Data Comes From

84-355 Democracy's Data | Session 1 — Election returns | Instructor copy — do not distribute

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All code in Part 7 is given to students. They are not writing it. What you grade is the interpretation underneath.

Every number in this key has been run against the actual data files.

The lab in one sentence

There is no official national election-returns file, so every county-level dataset in circulation was assembled by someone — and the four ways that assembly goes wrong (silent type coercion, missing third parties, name joins, and units that changed between years) are all visible in a single afternoon.

Students should leave unable to accept a county-level table at face value again.

Option A — Test the diagnosis

Verified output

```
cmp$h2_cty <- 100 * cmp$votes_dem / (cmp$votes_dem + cmp$votes_gop)
cmp$h2_st  <- 100 * cmp$harris    / (cmp$harris    + cmp$trump)
cmp$d2     <- round(cmp$h2_cty - cmp$h2_st, 2)

c(median_all_party = median(abs(cmp$dh)), median_two_party = median(abs(cmp$d2)))
```

```
#> median_all_party median_two_party
#>             0.02             0.00
```

```
c(agree_all_party  = sum(abs(cmp$dh) < 0.05), agree_two_party = sum(abs(cmp$d2) < 0.05))
```

```
#> agree_all_party agree_two_party
#>              30              49
```

```r
head(cmp[order(-abs(cmp$d2)), c("state_name","dh","d2","missing_pct")], 5)
```

```
#>    state_name   dh    d2 missing_pct
#> 36       Ohio 0.09  0.07        0.20
#> 2      Alaska 1.69 -0.06        3.92
#> 3     Arizona 0.00 -0.01        0.00
#> 48 Washington 0.40  0.01        0.70
#> 1     Alabama 0.13  0.00        0.39
```

The prediction holds, and decisively. Median gap falls from 0.02 to **0.00**. States agreeing within 0.05 points rise from **30 to 49 of 51**.

Only two states remain: **Ohio** (+0.07) and **Alaska** (−0.06).

What a good answer says

3/4: Reports that the gap collapsed and the agreeing-states count jumped, and concludes the denominator diagnosis was right.

4/4: Notices the two survivors are survivors for *different* reasons. Alaska's `missing_pct` is 3.92%, by far the worst in the country — it is the same disease, not yet fully cured, because Alaska's file is missing so much that even the two-party counts wobble. Ohio's `missing_pct` is 0.20%, essentially clean, so its residual is ordinary rounding in a source that publishes percentages to two decimals. Same symptom, two causes. A student who separates them is doing real diagnostic work.

4/4, the version to hope for: points out that this test *could have failed* and did not, and that this is what makes it evidence rather than decoration. If the county file's Harris counts had themselves been wrong, removing third parties would have changed nothing.

Common stumble: reporting "the medians are both about zero, so both methods are fine." That misses that the all-party median was already small — the informative number is the *count* of agreeing states, 30 versus 49.

Option B — Rank the states by data quality

Verified output

```r
head(cmp[order(-cmp$missing_pct), c("state_name","missing_pct","total_votes")], 8)
```

```
#>      state_name missing_pct total_votes
#> 2        Alaska        3.92      324097
#> 33     New York        2.17     8198094
#> 51      Wyoming        1.01      266353
#> 35 North Dakota        0.84      365059
#> 46      Vermont        0.82      366389
#> 21     Maryland        0.75     3015650
#> 14     Illinois        0.73     5592368
#> 48   Washington        0.70     3898835
```

```
round(cor(cmp$missing_pct, log(cmp$total_votes)), 3)
```

```
#> [1] -0.189
```

Worst: **Alaska 3.92%**, **New York 2.17%**, then Wyoming 1.01, North Dakota 0.84, Vermont 0.82, Maryland 0.75, Illinois 0.73, Washington 0.70.

Correlation with log(total votes) is −0.189 — essentially nothing.

What a good answer says

3/4: Names the worst states, reads the correlation correctly as near-zero, and concludes size does not explain incompleteness.

4/4: Resists inventing a pattern. Alaska and New York are outliers with different stories — Alaska because its whole reporting geography is unusual and its file is the most broken in the set, New York because it has a large and genuinely well-organised third-party ecosystem (Working Families, Conservative) whose ballots the compiler dropped. After those two, the tail is small and unstructured. "These are 51 separate administrative decisions, not a pattern" is the correct answer and should be rewarded, not treated as a failure to find something.

4/4 bonus: connects to the Bolts reading — the states that are hardest to track are not the ones you would predict, which is exactly the article's complaint. Whoever compiles these files is making choices nobody audits.

Common stumble: "Alaska is worst because it is rural/small." Wyoming is the only state that cast fewer votes than Alaska, and it is far less affected (1.01% against 3.92%); North Dakota and Vermont, which feel small, both cast *more* ballots than Alaska. Push back on this in feedback.

Option C — Where did the country move, and who is missing?

Verified output

```
head(sw[order(-sw$swing), c("state_name","county_name","swing")], 5)
```

```
#>      state_name     county_name swing
#> 2647      Texas Maverick County 14.07
#> 170  California Imperial County 12.89
#> 2725      Texas     Webb County 12.85
#> 1793   New York    Bronx County 11.27
#> 2699      Texas    Starr County 10.55
```

```
head(sw[order(sw$swing),  c("state_name","county_name","swing")], 5)
```

```
#>      state_name     county_name swing
#> 425     Georgia    Henry County -4.58
#> 2636      Texas   Loving County -4.17
#> 272    Colorado San Juan County -3.80
#> 224    Colorado  Chaffee County -3.48
#> 472     Georgia Rockdale County -3.31
```

```r
c(f2024 = nrow(cty), f2020 = nrow(old), matched = nrow(sw))
```

```
#>   f2024   f2020 matched
#>    3160    3152    3104
```

```r
setdiff(cty$state_name, sw$state_name)
```

```
#> [1] "Alaska"      "Connecticut"
```

```r
sw[sw$county_fips == "11001", c("county_name","r20","r24","swing")]
```

```
#>                county_name      r20      r24 swing
#> 283 District of Columbia 5.533046 5.280189 -0.25
```

Largest Republican swings: **Maverick TX 14.07**, **Imperial CA 12.89**, **Webb TX 12.85**, **Bronx NY 11.27**, **Starr TX 10.55**.

Largest Democratic swings: **Henry GA** −4.58, Loving TX −4.17, San Juan CO −3.80, Chaffee CO −3.48, Rockdale GA −3.31. Note the asymmetry — the Republican tail is 3.1 times longer than the Democratic one.

Join arithmetic: 3,160 units in 2024, 3,152 in 2020, **3,104 matched**.

Alaska and Connecticut are absent entirely — Alaska because the compiler silently renumbered its districts from `029xx` to `020xx`, Connecticut because the state replaced counties with planning regions and the codes share nothing.

The DC row: all of DC in 2020 (344,356 votes) against **Ward 1** in 2024 (40,463 votes, about 12% of the same city), both carrying code `11001`, producing a tidy −0.25 swing — a sign opposite to the median county's 1.53, which is exactly why nothing flags it.

What a good answer says

3/4: Describes the border-county and Bronx pattern, and names Alaska, Connecticut, and the DC row as problems.

4/4: Draws the inference that matters — the failures are *invisible in the output*. Two states vanish without an error message, and the DC row does not look wrong; it looks like every other row. A student who says "I would not have caught this if the lab hadn't told me, and that is the point" has understood the lab.

4/4, best version: asks what a national story built on this table would have claimed, and whether it would have been wrong. Mostly it would *not* have been — the Texas border swing is real and would survive. So the honest conclusion is uncomfortable: the file is broken in ways that happen not to change this particular headline, which is luck rather than method.

Common stumble: treating the DC row as a small error to be dropped. It is not about DC. It is about the fact that a join can succeed loudly and mean nothing.

Option D — Their own question

Grade the question, not the ambition. Re-running an existing chunk on a different state with a sharp observation is a 4. A new figure with no argument is a 2.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code. A chunk that failed to run is an office-hours problem, not a grade.
- **The distinctive skill this lab teaches is negative** — knowing what a dataset *cannot* support. Reward students who reach "this analysis is not possible with these files, and here is the check that shows it." Say so explicitly in feedback, because most students arrive believing a null answer is a failed one.
- **Watch for false pattern-finding**, especially in Option B. Confident stories about a correlation of -0.19 should be marked down, gently, with an explanation.

Anticipated questions

- *"Which file is right?"* — Neither is authoritative. The state file comes from a compilation of official returns and is closer to certified totals; the county file was scraped from Fox News's results pages, as its own README admits. The useful answer is that "right" depends on the question, and the compiler's choices are documented nowhere except in the discrepancies.
- *"Why not just use official data?"* — There is no national official source. That is the Bolts reading's entire argument, and this is the week to make the connection explicit.
- *"Did anyone publish anything wrong because of this?"* — Almost certainly, and quietly. Good Option D territory.
- *"Is the DC thing a bug in the lab?"* — No. It is in the source data, and it is the most important thing in the lab.

Connections forward

- **Data provenance** recurs in **Session 2** — the Census is the one place the US does have an official national enumeration, and that session is about what it costs to build one.
- **Geographic units changing under you** is the modifiable areal unit problem, formally introduced in the `census-geography/` lab and again in **Session 14**. Connecticut is the live example; refer back to it by name.
- **Ecological inference** in **Session 14** depends entirely on trusting a county-to-precinct join. Students who did Option C will understand why that assumption deserves scrutiny.
- The **Bolts** article assigned this week is the argument; this lab is the demonstration. Ask on Tuesday who is harmed when election data is this hard to assemble — the answer, per the article, is voting-rights litigants, and that points straight at **Session 3 (§203)**.